Generating SSH Keys for GCE

https://gemini.google.com/app/965d853d584a53f9

User prompt: Google Cloud Engine SSH key generate

Response: To generate an SSH key pair and add it to Google Cloud Compute Engine (GCE), you create an ED25519 or RSA key on your local terminal, format the public key for GCP, and apply it to your project or instance.

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1 Generate the Key Pair

Terminal / PowerShell

Run `ssh-keygen` using the ED25519 algorithm. GCP requires a comment field at the end of the public key formatted as `USERNAME`.

```
ssh-keygen -t ed25519 -C "USERNAME" -f ~/.ssh/gcp_key
```

Replace `USERNAME` with your target Linux username on the VM.

2 Format the Public Key

Required by GCP

Display your public key content:

```
cat ~/.ssh/gcp_key.pub
```

Ensure the key output follows this exact standard format: `ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAI... USERNAME`

3 Add Key to Google Cloud

Console or gcloud CLI

Option A (GCP Console):

1. Go to **Compute Engine** > **Metadata** (for all VMs) or select a specific **VM instance** > click **Edit** (for a single VM).
2. Scroll to **SSH Keys** and click **Add Item**.
3. Paste the contents of `gcp_key.pub` into the text box and click **Save**.

Option B (gcloud CLI):

```
gcloud compute project-info add-metadata \
  --metadata ssh-keys="USERNAME:$(cat ~/.ssh/gcp_key.pub)"
```

4 Connect to Your Instance

Verification

Connect via terminal using your private key and the external IP address of your instance:

```
ssh -i ~/.ssh/gcp_key USERNAME@EXTERNAL_IP
```

To verify success: running `whoami` after connecting should return your specified `USERNAME`.
